FINAL PROJECT

SECR3443 CRYPTOGRAPHY

SEMESTER I, 2024/25

Section: 02

Lecturer: Ms Rashidah Kadir

GROUP MEMBERS:

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Website Link: https://cryptosphere-project.vercel.app/

Video Link:https://youtu.be/KmH27rc3KC8?si=1DE_5rOxEPU2QMat

Coding Link: Cryptosphere-Project-Code

Project Report: CryptoSphere Interactive

1. Problem Statement

Secondary school students (ages 13–17) often find cryptography concepts like "integrity" or "transposition" difficult to visualize. Traditional static textbooks cannot provide the immediate, personalized feedback required when a student makes a logic error. This results in a "learning gap" where students may follow instructions but fail to understand the underlying cryptographic principles.

2. Target Users

- **Primary:** Secondary school students (Ages 13–17).
- **User Profile:** Digital natives with no prior background in security, requiring high engagement and simplified explanations.

3. Learning Outcomes

1. **Execute** Caesar/Substitution and Transposition ciphers through interactive modules.
2. **Demonstrate** the concept of Data Integrity using Hashing functions.
3. **Reflect** on the accuracy of AI-generated feedback and identify potential AI risks.

4. App/Game Concept: "CryptoSphere"

CryptoSphere is a high-performance, reactive web application built as a specialized training simulator for secondary school students. By moving from a Java backend to a **full-stack TypeScript/React** approach on Vite, the app provides instantaneous feedback, which is crucial for maintaining the engagement of 13–17-year-olds.

Level 1: The Shift Key (Caesar/Substitution)

- **The Mechanic:** Using a custom-built slider component (styled with **Tailwind CSS**), students interact with a dynamic alphabet array. As the student adjusts the slider, the TypeScript logic re-renders the ciphertext in real-time.
- **Goal:** Decrypt an "Intercepted Memo" to gain entry into the virtual forensic lab.

Level 2: The Scrambled Grid (Transposition)

- **The Mechanic:** A drag-and-drop grid interface. Students must take a linear string of characters and "re-structure" them into a matrix. The **React state** manages the position of each character block.
- **Goal:** Reorganize the "Scrambled Data Packets" to reveal a hidden admin password.

Level 3: The Integrity Scan (Hashing)

- **The Mechanic:** A "Digital Fingerprint" scanner. Students are presented with two input fields. As they type or modify files, a hashing algorithm (implemented in **TypeScript**) generates a real-time SHA-256 string.
- **Goal:** Identify a "Tampered File" by noticing that even a single-character change results in a completely different hash value.

5. Tools Used (Technical Stack)

Framework: Vite + React (For a fast, reactive single-page application).

Language: TypeScript (To ensure type-safe cryptographic logic and state management).

UI/UX: shadcn-ui + Tailwind CSS (To build a professional, accessible, and "dark-mode" forensic-themed interface).

AI Engine: OpenAI API / Vercel AI SDK (To power the adaptive hints and feedback).

Testing: Manual testing with **7–11 students** using the mandatory **7-item Likert scale** questionnaire from the project brief.

6. Game Flow

The game is structured as a single-page application (SPA) that uses a state-driven progression. This ensures a seamless transition for the student without page reloads.

Phase 1: Entry & Mission Briefing

- **Interface:** A clean, dark-themed dashboard built with **shadcn-ui** cards.
- **Flow:** The student is greeted as a "Forensic Analyst." The UI displays three locked modules. They must complete them in order to ensure a logical "scaffolding" of knowledge.

Phase 2: Active Investigation (Levels 1-3)

- **Level 1 (Caesar/Substitution):** The student interacts with a "Cipher Wheel" component. They input a shift value and watch the plaintext transform into ciphertext.
- **Level 2 (Transposition):** A grid-based puzzle. The student must arrange a jumbled string into the correct column-row matrix.
- **Level 3 (Hashing):** A "Data Integrity Scanner." The student compares two large files. They must identify a single-character change that caused a "Hash Mismatch."

Phase 3: The AI Debriefing

- **Flow:** After each level, a **feedback modal** appears. If the student struggled (multiple incorrect attempts tracked in TypeScript state), the **Adaptive Hint** feature kicks in.
- **Completion:** Once all three hashes are verified, the student submits their final "Forensic Report" (an explanation of the concepts), which the AI grades with **Automated Feedback**.

7. Cryptography Explanations

To meet the project objective of "translating complex ideas into simple explanations," the app uses the following logic:

Module 1: Caesar & Substitution (Confidentiality)

- **The Concept:** Explained as a "Mechanical Shift."
- **The Logic:** If you share a secret "Key" (e.g., Number 5), you move every letter 5 steps forward. To read it, the receiver moves 5 steps back.

- **Educational Goal:** To show that even if a message is intercepted, it is unreadable without the key.

Module 2: Transposition (Diffusion)

- **The Concept:** Explained as "Scrambling the Map."
- **The Logic:** We don't change the letters; we change their positions. By writing a message in a 3x3 grid and reading it vertically, the sentence "THE CAT SAT" becomes "TCT HAA ESA."
- **Educational Goal:** To demonstrate that the structure of data can be hidden without changing the characters themselves.

Module 3: Hashing & Integrity (The Digital Fingerprint)

- **The Concept:** Explained as a "One-Way Seal."
- **The Logic:** A Hash function takes any amount of data and turns it into a short, unique string of characters. You cannot turn the hash back into the message.
- **The "Integrity" Lesson:** If a hacker changes a single bit of data, the "Fingerprint" (Hash) breaks. If the Fingerprints don't match, the data is no longer "Integer" (whole/untainted)

8. AI Integration (Core Features)

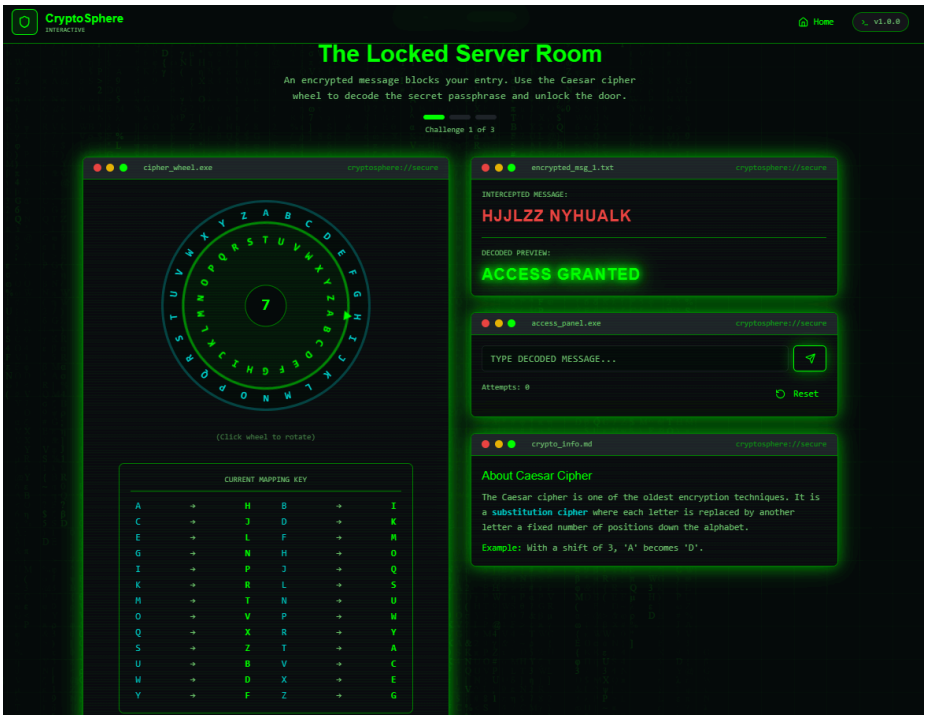
Feature 1: Adaptive Hints: The **TypeScript** code includes a counter for errors.

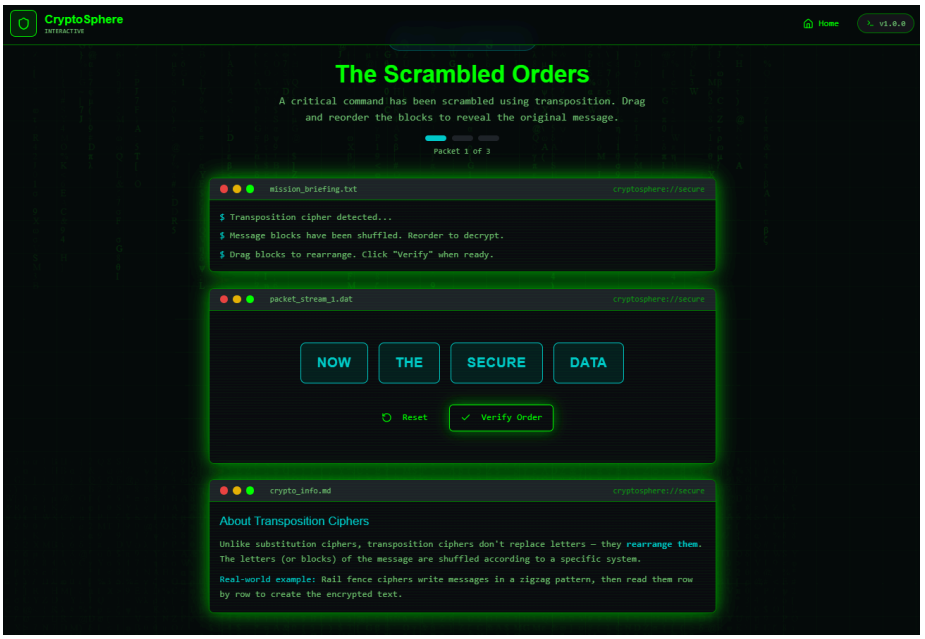
- *Logic: if (attempts > 3) { triggerAIHint(currentLevel, userMistake);}*
- *Result:* The student receives a hint like: *"It looks like you are trying to decrypt using the wrong shift. Remember: to decrypt, you must move the wheel BACKWARDS by the key amount."*

Feature 2: Automated Feedback: Using **React state**, the student's written explanation of "Integrity" is captured and sent to the AI.

- *Result:* Instead of just "Correct," the AI responds: *"Excellent explanation! You correctly identified that Hashing is a one-way process. This is exactly why it's used to check for file tampering."*

9. Screenshots





packet_stream_1.dat

cryptosphere://secure

NOW SECURE THE DATA

Reset

Verify Order

Try This

Incorrect order. This is a command. What is the action verb?

CryptoSphere
INTERACTIVE

Home

v1.0.0

LEVEL 3: HASHING

The Digital Fingerprint

Explore how cryptographic hashing creates unique "fingerprints" for data. Even a single bit change alters the entire fingerprint.

Mission: The Avalanche Effect

input_buffer.txt

cryptosphere://secure

Hashes change drastically with tiny edits. Modify the text 3 times to observe this.

CLUE: Click to reveal clue...

The quick brown fox jumps over the lazy dog.

Length: 44 chars

Reset

sha256_processor.exe

cryptosphere://secure

LIVE HASH OUTPUT:

ef537f25c895bfa782526529a9b63d97aa631564d5d789c2b765448c8635fb6c

mission_output.log

cryptosphere://secure

STATUS LOG:

Modifications: 0/3

Verify Experiment

>_ HASH_COMPARISON.LOG

ORIGINAL: ef537f25c895bfa782526529a9b63d97aa631564d5d789c2b765448c8635fb6c

CURRENT: ef537f25c895bfa782526529a9b63d97aa631564d5d789c2b765448c8635fb6c

crypto_info.md

cryptosphere://secure

About Cryptographic Hashing

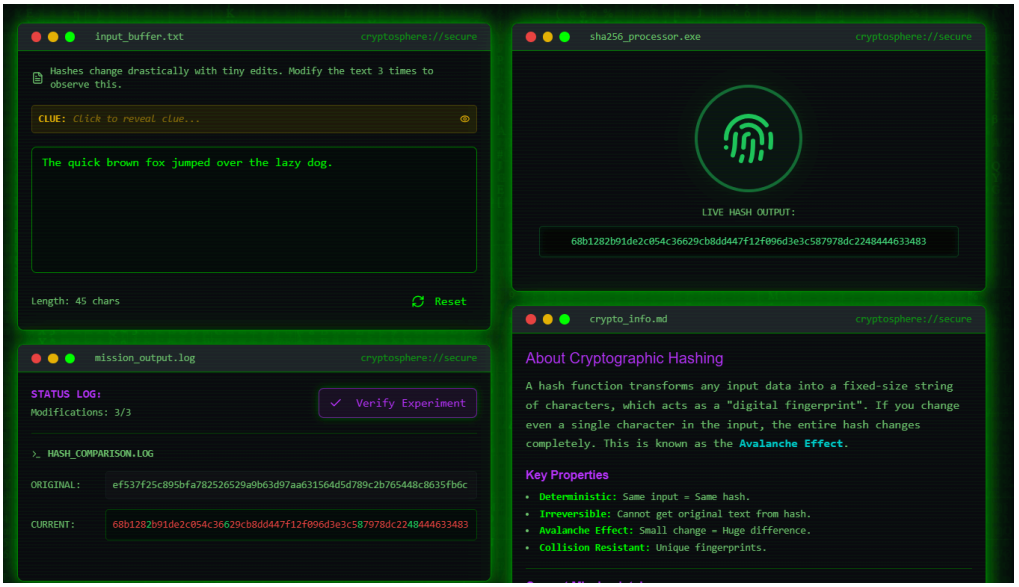
A hash function transforms any input data into a fixed-size string of characters, which acts as a "digital fingerprint". If you change even a single character in the input, the entire hash changes completely. This is known as the **Avalanche Effect**.

Key Properties

- **Deterministic:** Same input = Same hash.
- **Irreversible:** Cannot get original text from hash.
- **Avalanche Effect:** Small change = Huge difference.
- **Collision Resistant:** Unique fingerprints.

Current Mission Intel:

Notice how a tiny change (like A -> a) completely changes the "fingerprint"? This is called the Avalanche Effect.


input_buffer.txt
cryptosphere://secure
Hashes change drastically with tiny edits. Modify the text 3 times to observe this.
CLUE: Click to reveal clue...
The quick brown fox jumped over the lazy dog.
Length: 45 chars
Reset
sha256_processor.exe
cryptosphere://secure
LIVE HASH OUTPUT:
68b1282b91de2c054c36629cb8dd447f12f096d3e3c587978dc2248444633483
mission_output.log
cryptosphere://secure
STATUS LOG:
Modifications: 3/3
Verify Experiment
>_ HASH_COMPARISON.LOG
ORIGINAL: ef537f25c895bfa782526529a9b63d97aa631564d5d789c2b765448c8635fb6c
CURRENT: 68b1282b91de2c054c36629cb8dd447f12f096d3e3c587978dc2248444633483
crypto_info.md
cryptosphere://secure
About Cryptographic Hashing
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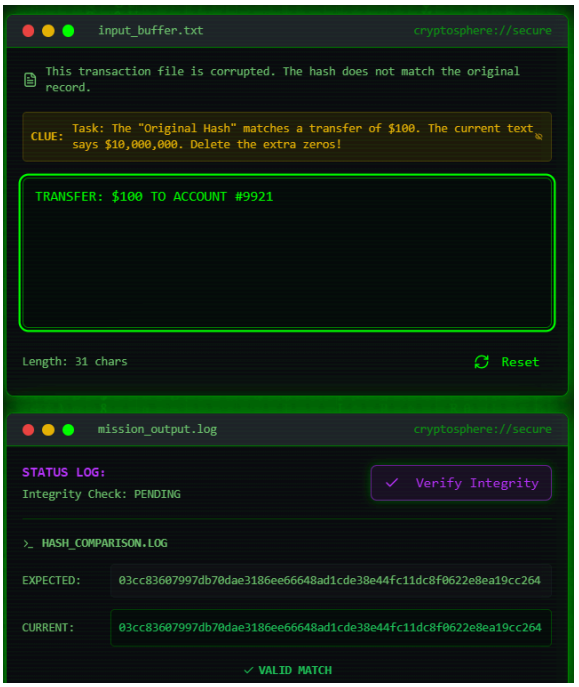
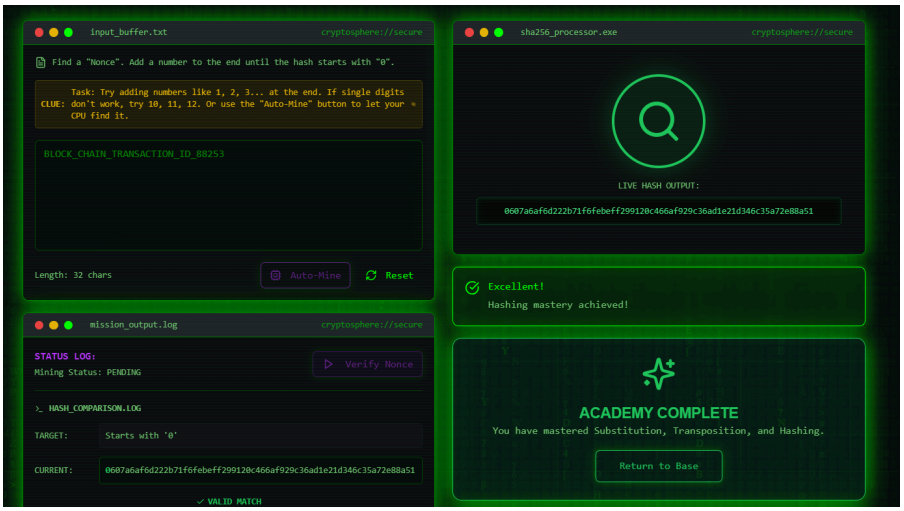
Key Properties
Deterministic: Same input = Same hash.
Irreversible: Cannot get original text from hash.
Avalanche Effect: Small change = Huge difference.
Collision Resistant: Unique fingerprints.

input_buffer.txt
cryptosphere://secure
This transaction file is corrupted. The hash does not match the original record.
CLUE: Task: The "Original Hash" matches a transfer of $100. The current text says $10,000,000. Delete the extra zeros!
TRANSFER: $10,000,000 TO ACCOUNT #9921
Length: 38 chars
Reset
mission_output.log
cryptosphere://secure
STATUS LOG:
Integrity Check: PENDING
Verify Integrity
>_ HASH_COMPARISON.LOG
EXPECTED: 03cc83607997db70dae3186ee66648ad1cde38e44fc11dc8f0622e8ea19cc264
CURRENT: 73a0fefb73acca1e54d80af6b02493da64ba072b2aa95b9271b6d349c368a1f9


input_buffer.txt
cryptosphere://secure
This transaction file is corrupted. The hash does not match the original record.
CLUE: Task: The "Original Hash" matches a transfer of $100. The current text says $10,000,000. Delete the extra zeros!
TRANSFER: $100 TO ACCOUNT #9921
Length: 31 chars
Reset
mission_output.log
cryptosphere://secure
STATUS LOG:
Integrity Check: PENDING
Verify Integrity
>_ HASH_COMPARISON.LOG
EXPECTED: 03cc83607997db70dae3186ee66648ad1cde38e44fc11dc8f0622e8ea19cc264
CURRENT: 03cc83607997db70dae3186ee66648ad1cde38e44fc11dc8f0622e8ea19cc264
VALID MATCH

input_buffer.txt
cryptosphere://secure
Find a "Nonce". Add a number to the end until the hash starts with "0".
CLUE: Task: Try adding numbers like 1, 2, 3... at the end. If single digits don't work, try 10, 11, 12. Or use the "Auto-Mine" button to let your CPU find it.
BLOCK_CHAIN_TRANSACTION_ID_882
Length: 30 chars
Auto-Mine
Reset
mission_output.log
cryptosphere://secure
STATUS LOG:
Mining Status: PENDING
Verify Nonce
>_ HASH_COMPARISON.LOG
TARGET: Starts with '0'
CURRENT: d172f9035a77798c3ad503fc528a8774e64ccd30d6dcd6b4650ba6391f5383f5


input_buffer.txt
cryptosphere://secure
Find a "Nonce". Add a number to the end until the hash starts with "0".
CLUE: Task: Try adding numbers like 1, 2, 3... at the end. If single digits don't work, try 10, 11, 12. Or use the "Auto-Mine" button to let your CPU find it.
BLOCK_CHAIN_TRANSACTION_ID_88253
Length: 32 chars
Auto-Mine
Reset
mission_output.log
cryptosphere://secure
STATUS LOG:
Mining Status: PENDING
Verify Nonce
>_ HASH_COMPARISON.LOG
TARGET: Starts with '0'
CURRENT: 0607a6af6d222b71f6febeff299120c466af929c36ad1e21d346c35a72e88a51
✓ VALID MATCH
sha256_processor.exe
cryptosphere://secure
LIVE HASH OUTPUT:
0607a6af6d222b71f6febeff299120c466af929c36ad1e21d346c35a72e88a51
Excellent!
Hashing mastery achieved!
ACADEMY COMPLETE
You have mastered Substitution, Transposition, and Hashing.
Return to Base

10. User Testing

10.1 Methodology

To evaluate the effectiveness of our website CryptoSphere, we have conducted a user testing with **11 secondary school students (aged 13-17)**. Participants completed all three levels and then submitted feedback via a Google Form.

10.2 Feedback Questionnaire Design

The questionnaire consisted of 6 mandatory questions that use Likert-scale (1 = Strongly Disagree to 5 = Strongly Agree) and 2 optional open-ended questions.

Questions:

1. The application/game is easy to use and understand.
2. The instructions and explanations are clear.
3. The cryptography concepts are easy to understand after using the app/game.
4. The activities or games are engaging and enjoyable.
5. The AI feature(s) provide helpful guidance or feedback.
6. I would recommend this app/game to other students.
7. What did you like most about the app/game?
8. What improvements would you suggest?

10.3 Quantitative Results (Q1-Q6)

Question	Criteria	Average Score	Interpretation
1	Usability	3.1	Students found it hard to use/do the challenges
2	Clarity	3.0	Instructions required re-reading and hard to understand
3	Educational Value	4.2	Students successfully grasped the core cryptography concepts
4	Engagement	4.6	The gamified "Cyberpunk" theme keeps students interested.
5	Ai Helpfulness	4.6	The adaptive hints were extremely effective
6	Recommendation	4.3	Students willing to recommend to others

Analysis of Findings: Distinct contrast between Engagement and Clarity.

- **Strengths:** Highest scores were 4.6 for both **engagement** and **AI Helpfulness.** The core objective of the rule-based AI system successfully guided students through the challenges and the web design of "cyberpunk" theme kept them interested.

- **Weakness:** The lower score in **Usability (3.1)** and **Clarity (3.0)** indicate that while the game was fun it had a steep learning curve. The design, while immersive, may have resulted in making the instructions harder to read and understand for some users.

10.4 Quanlitative Results (Q7-Q8)

Analysis of Findings:

Positive Feedback (Q7)

- Many respondents liked the visual aspects and the design of the website.
- Students like the information content and the engaging activities.

Constructive Feedback (Q8)

- Students want the instructions and tasks to be clearer and easier to understand.
- Students want more challenges.

10.5 Conclusion

The user testing confirms that CryptoSphere is highly successful as an educational tool. The lower scores in usability and clarity suggest that the instruction should be simpler to understand. However, since the recommendation score (4.3) remains high, users felt it was fun rather than confused.

Future Improvement:

1. Add a tutorial level for each category of challenges.

10.5 User Feedback Link

https://forms.gle/DharDaAGrzNsfQwN6

11. Ethics, Safety, and Responsibility

- **Student Data Privacy:** We adhere to a strict "No Data Collection" policy. The website does not require any login and signup. The user feedback forms also only focussed on questions and do not ask any personal user information like name, age, etc.

- **AI Limitations:** To prevent AI hallucinations, we use **Rule-Based Expert System** instead of Generative AI like ChatGPT. This ensures the hints that the user needs are based on the user's progress and 100% accurate, avoiding the risk of teaching false concepts.

- **Preventing Misuse:** Instead of framing hackers as the bad people and using offensive hacking, we frame cryptography as a defensive tool for digital detectives. This **White Hat** context ensures students understand cryptography for ethical purposes.

- **Age-Appropriate Content:** We replaced all complex mathematical concepts with interactive visual metaphors like rotating wheels, colour-changing hashes to make the concept easily understood by people in the 13-17 age group.

12. Video Presentation

Link:https://youtu.be/KmH27rc3KC8?si=1DE_5rOxEPU2QMat

13. Individual Reflection

AVIDIAN DIPESH SIVA

Contribution	• Core Technical Development: Led the primary programming of the CryptoSphere web application. • Interactive Module Design: Developed the custom slider component for Caesar ciphers and the drag-and-drop grid interface for transposition challenges. • AI Logic Integration: Programmed the TypeScript error-counting system that triggers adaptive hints to provide student scaffolding during logic errors.
Skills Learned	• Full-Stack SPA Development: Mastered building fast, single-page applications (SPA) using TypeScript to ensure type-safe cryptographic logic. • Deterministic AI Design: Learned to implement rule-based expert systems for educational feedback to prevent AI hallucinations and ensure 100% accuracy. • Practical Cryptography: Gained experience translating abstract theories like SHA-256 and matrices into functional, real-time visual metaphors.
Challenges Faced	• Visualizing Integrity: Converting the abstract concept of "Integrity" into a visual mechanic; solved by creating the live SHA-256 "Digital Fingerprint" scanner. • Complex State Management: Managing user progression and unlocking locked modules sequentially without page reloads using React state.

MERVYN BEE ZHENG CHENG

1. Contribution

   My contribution was centered on the strategic planning and final delivery of the project. I acted as the bridge between the technical development and the educational requirements of the brief.

   Specifically, I:

   - **Ideation & Conceptualization:** I proposed the initial game concept, "The Digital Detective" ensuring it covered the three required concepts (Caesar, Transposition, and Hashing) and integrated AI in a meaningful way through adaptive hints and feedback.
   - **Project Proposal:** I authored the full project proposal, defining the learning outcomes, target user profile (ages 13-17), and the justification for using the **Vite/React/TypeScript** stack.
   - **Documentation:** I compiled the 8-page final Design Documentation, including the system architecture, AI design logic, and the analysis of user testing results.
   - **Video Presentation:** I took part in the **3-5 minute video presentation** for YouTube, ensuring it clearly explained our cryptography concepts and showcased a live demo of our AI integration.
2. Skills learned
   - **Educational Communication:** I learned how to translate dense cryptographic theory into "simplified" explanations that satisfy the project's goal of being age-appropriate for teenagers.
   - **User Experience (UX) Evaluation:** Through managing the user testing, I learned how to interpret Likert-scale data to determine if our AI features were actually helping students or causing more confusion.
   - **Technical Writing & Documentation:** I improved my ability to document a technical stack (Vite, React, shadcn-ui) and explain the "Responsible AI" measures we took to prevent AI hallucinations.
3. Challenges faced
   - **Aligning AI with Pedagogy:** One of the main challenges was ensuring the AI hints didn't simply "give away the answer." I had to work closely with the developer to refine the prompts so that they provided **Scaffolding** (small steps) rather than full solutions.
   - **Time Management:** With the deadline of **20 Jan 2026**, balancing the final video editing with the completion of the individual reflections and the final report required strict scheduling and constant communication with my team.

MUHAMMAD AZIIDAN BIN MUHD AZLAN NG

Contribution

My primary contribution was the technical execution and software development of the CryptoSphere web application. I was responsible for helping Avidian to finalize the web application and host it so that users can access it easily.

- **Core Development:** I built the application using React framework with TypeScript and Vite, creating simple pages that allows rapid development and smooth transitions between levels without page reloads. Specifically, more tasks for each category rather than one, AI information page and appreciation page with feedback link.
- **Documentation:** I fix all the formats of existing documentation so that it is aligned throughout the document. I am also responsible for 10. User Testing and 11. Ethics, Safety, and Responsibility.
- **User Testing:** I make the Google Forms for the user feedback and responsible to distribute them to target users which are secondary school students.

Skills Learned

- **Responsible AI Design:** I learned the distinction between Generative AI and Deterministic AI. Building this project taught me how to create AI feedback systems that are safe, accurate and no hallucinations which is very important for educational software.
- **Educational Communication:** I learn how to simplify dense cryptography theory to simpler terms that fit the learning curve for secondary school students age 13-17.

Challenges Faced

- **Defining AI to use:** A major issue was to meet the project requirement for "AI Integration" without using API like Gemini AI (due to cost concerns). I had to discuss and design an intelligent system that detect user specific error patterns and serving and serving custom feedback, ensuring it met "Adaptive Hints" criteria.
- **Deployment and Hosting:** Deploying the website to Vercel requires a specific github repository and I learned how to commit huge files of initial codes that I got from Avidian. Then proceed to hosting it using Vercel which is free.

TEE SOON KIET

Contribution

I acted as the bridge between technical development and educational requirements, ensuring the project met the secondary school curriculum standards.

- **Ideation & Conceptualization:** Proposed the "CryptoSphere" simulator and the transition from a Java backend to a reactive TypeScript/React frontend for improved performance.
- **Pedagogical Logic:** Designed simplified cryptographic metaphors, such as the "Digital Fingerprint" for Hashing, to bridge the learning gap for students.
- **Video Presentation:** Co-produced the final presentation, demonstrating how the first challenge of our site work and what it teaches.

Skills Learned

- **Educational Communication:** Mastered translating complex theories (SHA-256/Transposition) into age-appropriate language for teenagers.
- **UX Evaluation:** Gained experience analysing Likert scale data from user testing to assess the impact of interactive components on student engagement.
- **Technical Writing & AI Governance:** Improved documentation skills regarding technical stacks and "Responsible AI" measures to prevent AI hallucinations.

Challenges Faced

- **Balancing AI Assistance:** Refined prompt logic to ensure AI provided "Scaffolding" (hints) rather than direct answers to maintain the educational challenge.
- **Visualizing Abstract Concepts:** Overcame the difficulty of teaching "Integrity" by creating a real-time visual scanner where SHA-256 strings react instantly to data changes.
- **Time Management:** Successfully coordinated technical documentation, Likert-scale analysis, and video production to meet the strict 20 Jan 2026 deadline.